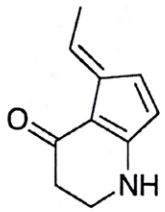
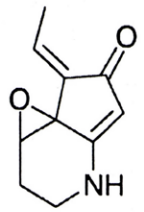
Organic Cumulative Exam Friday, February 3, 2017

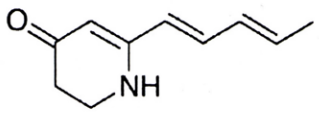
The four structures below were isolated from Streptomyces strains of bacteria (*J. Nat. Prod.* **2000**, *63*, 1258-1260. Four data sets are given on the following page. Correlate each structure with the appropriate data set. In order to receive credit, you must briefly describe the distinguishing spectral features of each compound and support it by interpreting the relevant data from the data sets.



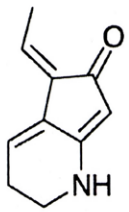
A



B



C



D

data set #1
white solid, mp 133 C; Rf 0.27 (CHCl3-MeOH, 9:1), 0.28 (EtOAc-cyclohexane-MeOH, 10:5:1); color reaction with Ehrlich's reagent: pink; UV (MeOH) max (log) 239 (4.19), 310 (4.37) nm; (MeOH + HCl) 220 (4.18), 315 (4.34) nm; CD (MeOH) () 240 (-1.50), 254, (+0.26), 290 (-5.50), 320 (+5.48) nm; IR (KBr) max 1654, 1585 cm-1; 1H NMR (300 MHz, acetone-d6) 7.32 (1H, br s), 5.69 (1H, qd, J = 7.3, 0.7 Hz), 5.18 (1H, s), 3.95 (1H, d, J = 2.7 Hz), 3.35 (1H, ddd, J = 12.3, 12.3, 4.0 Hz), 3.18 (1H, dddm, J = 12.3, 5.5, 5.5 Hz), 2.28 (1H, dm, J = 15.0 Hz), 2.15 (3H, d, J = 7.3 Hz), 2.13-2.06 (1H, m); 13C NMR (125.7 MHz, acetone-d6) 191.7 (s), 166.6 (s), 133.4 (s), 128.2 (d), 105.9 (d), 60.1 (s), 59.6 (d), 37.6 (t), 26.2 (t), 13.6 (q);

data set #2
colorless oil, Rf 0.32 (CHCl3-MeOH, 9:1), 0.36 (EtOAc-cyclohexane-MeOH, 10:5:1); color reaction with Ehrlich's reagent, violet; UV (MeOH) max (log) 262 (4.14), 315 (3.99) nm; (MeOH + HCl) 262 (4.04), 279 (4.02), 327 (3.84) nm;
IR (KBr) max 1665, 1635, 1628, 1580 cm-1; 1H NMR (500 MHz,CD3OD)
6.31 (1H, td, J = 4.5, 1.3 Hz), 6.25 (1H, q, J = 7.5 Hz), 5.17 (1H, s br), 3.36 (2H, t, J = 7.0 Hz), 2.47 (2H, td, J = 7.0, 4.5 Hz), 2.22 (3H, d, J = 7.5 Hz); 13C NMR (125.7 MHz, CD3OD) 195.1 (s), 165.7 (s), 132.4 (s), 132.3 (s), 126.7 (d), 119.2 (d), 102.4 (d), 39.9 (t), 24.0 (t), 13.6 (q);

data set #3
yellow oil, Rf 0.30 (CHCl3-MeOH, 9:1), 0.29 (EtOAc-cyclohexane-MeOH, 10:5:1); color reaction with Ehrlich's reagent, gray-violet; UV (MeOH) max (log) 286 (3.76), 368 (3.39) nm; (MeOH + HCl) 224 (3.58), 276 (3.71), 314 (3.57) nm;
IR (KBr) max 1613, 1571 cm-1; 1H NMR (500 MHz, CD3OD) 7.20 (1H, dm, J = 5.5 Hz), 6.86 (1H, qm, J = 8.0 Hz), 6.28 (1H, dm, J = 5.5 Hz), 3.59 (2H, t, J = 8.0 Hz), 2.38 (2H, t, J = 8.0 Hz), 2.04 (3H, d, J = 8.0 Hz);13C NMR (125.7 MHz, CD3OD) 187.0 (s), 169.0 (s), 141.0 (s), 136.2 (d), 131.0 (d), 124.2 (d), 103.8 (s), 43.3 (t), 36.8 (t), 15.8 (q);

data set #4
yellow solid, mp 105-109 C; Rf 0.28 (CHCl3-MeOH, 9:1) 0.24 (EtOAc-cyclohexane-MeOH, 10:5:1); color reaction with Ehrlich's reagent, violet; UV (MeOH) max (log) 288 (4.34), 361 (4.11) nm; (MeOH + HCl) 215 (3.91), 314 (4.43) nm;
IR (KBr) max 1641, 1592, 1524 cm-1; 1H NMR (500 MHz, CD3OD) 6.93 (1H, dd, J = 15.5, 10.3 Hz), 6.23 (1H, ddq, J = 15.0, 10.3, 1.0 Hz), 6.09 (1H, dq, J = 15.0, 7.0 Hz), 5.92 (1H, d, J = 15.5 Hz), 5.11 (1H, s), 3.54 (2H, t, J = 7.7 Hz), 2.37 (2H, t, J = 7.7 Hz), 1.84 (3H, dd, J = 7.0, 1.0 Hz); 13C NMR (125.7 MHz, CD3OD) 194.8 (s), 163.2 (s), 138.6 (d), 138.1 (d), 131.9 (d), 124.3 (d), 96.8 (d), 41.9 (t), 35.6 (t), 18.7 (q);